

Akshit Rampershad

Turning enterprise operations into scalable AI-native systems

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EXPERIENCE

AI Engineer | Cambrian Lab, San Francisco, CA

May 2026 – Present

- Architected an **end-to-end PCN-MPN matching system using a 4-leg hybrid retriever** (exact, fuzzy, prefix, vector) feeding a bounded **LLM ranking pass**, with a short-circuit exact-match mechanism that resolves deterministic hits at 0 LLM cost.
- Optimized **vector search to a flat 0.36s match latency** on a 50K+ row corpus, embedding 4 manufacturing fields with **OpenAI text-embedding-3-small 1536-d** and **routing HNSW top-K candidates** through a 3-band score+margin policy.
- Validated hybrid-retrieval candidate matches using a 3-band score+margin confidence policy. **reducing mismatched-part risk in PCNs, achieving 0 false auto confirmations across 81 validation runs** and automating unambiguous matches.
- Eliminated manual assembly and finished good tracing by building a **recursive "where-used" BOM engine with parent-traversal graph algorithms and cycle-detection guards**, fully automating downstream impact analysis for matched part.
- Built PcnAutoIntake, an **LLM orchestration pipeline** automating 29+ daily supplier change intake via structured extraction, entity resolution and confidence gated human reviews, **cutting PCN analysis from 2hr to 10sec/notice, saving 19hrs/day**.

Data Engineer / AI Solutions Delivery | 22nd Century Technology Inc. McLean, VA

Sept 2025 – Mar 2026

- Spearheaded the **end-to-end technical delivery of a secure enterprise RAG platform** to accelerate multi-million dollar federal proposal development cycles and intelligence generation workloads across 20K+ SharePoint knowledge corpus.
- Built a **hybrid ingestion engine using Python, PySpark, and Graph API to index a 20K+ SharePoint corpus into a 1,024-dim Qdrant vector database**, using BGE-Large-EN-v1.5 embeddings for section-aware retrieval.
- Designed a **LlamaIndex retrieval pipeline narrowing 30 candidates to 8 through five-field metadata filtering and Cross-Encoder re-ranking**, feeding LangChain prompt orchestration to return RFP-scoped drafts sourced from prior wins.
- Managed project dependencies and technical ambiguity for 4 cross-functional teams through Jira, **driving stakeholder collaboration across AI, engineering, and proposal teams** to deliver structured RAG features across 8 continuous sprints.
- Developed AI adoption runbooks and **led technical enablement sessions with business stakeholders and 14 proposals writers**, improving organizational data literacy and scaling RAG application adoption by 45% across business teams.

Data Engineer / AI Practitioner | Surge Infotech, Boston, MA

Sept 2024 – Sept 2025

- Engineered **high-throughput spatial data ingestion pipelines and an ML site-ranking engine** using Databricks AutoML and MLflow, surfacing optimal urban development parcels for real estate clients with a 92% prediction accuracy.
- Built **FastAPI data extraction services to ingest 50M+ parcel and zoning records into AWS S3**, leveraging Databricks Auto Loader for incremental file discovery and schema inference and DLT for governed ingestion and automated quality checks.
- Orchestrated multi-stage Medallion pipelines (Bronze/Silver/Gold) using Databricks LakeFlow and DLT, **processing 50M+ spatial and zoning records with automated schema evolution and real-time quality validation**.
- Leveraged **OpenAI GPT-4 APIs to build an automated intelligence engine that synthesized complex pipeline metrics into executive-ready business reports**, accelerating C-suite strategic planning cycles by 75%.
- Presented smart city market adoption trends to executive leadership, aligning foundational model capabilities with commercial demands to **secure executive buy-in for a multi-quarter AI engineering product roadmap**.

Business Data Analyst | UKG, India

Oct 2020 – Aug 2022

- Led business and technical requirements analysis for UKG TALK post-acquisition, **identifying key product gaps and shaping our strategic data backed product roadmaps**, successfully supported 10,000+ active enterprise users worldwide.
- Detected on-boarding drop-offs using **SQL funnel analysis and Python cohort segmentation**, executing process optimization techniques that boosted premium **feature monthly conversions rate by 29%** and **trial to paid rate by 25%**.
- Developed **JIRA defect tracking dashboards monitoring production/staging issues**, enabling 4 cross-functional teams to unblock dependencies and critical bugs, **resulting a 43% improvement in phased sign-offs with 94% sprint delivery rate**.
- Designed **interactive Tableau and Power BI dashboards using advanced parameters and calculated fields** to report real-time campaign ROI and segment conversion metrics to non-technical stakeholders during executive quarterly reviews.
- Led **cross-functional UAT loops, post-production checks, and product marketing launch documentation** to control critical integration friction for high-value Fortune 500 clients, **boosting platform adoption rates by 2X within 12 months**.

EDUCATION

- **MS in Business Analytics and Information Systems**, University of South Florida, Tampa, Florida **May 2024**
- **BS in Computer Science and Engineering**, Gandhi Institute of Technology and Management, India **July 2020**

SKILLS

- **Agentic AI & LLM Orchestration**: Anthropic Claude API, OpenAI GPT-4, Gemini-3.7-flash, LangChain, LlamaIndex, MCP, Prompt Engineering, Data Extraction, Tool Orchestration, Multi-System Agent Workflows, Autonomous Agentic Engines.
- **RAG & Retrieval**: Hybrid Retrieval (BM25/dense/RRF), Qdrant, Vector Search (HNSW/ANN), 2Stage Retrieval, Embedding (text-embedding3small/g-embedding001/BGELarge), Entity Resolution, Cross-Encoder Rank, Payload Indexing, Chunking.
- **AI Evaluation & Reliability**: Confidence Calibration, Threshold/Margin Policy, Validation Harnesses, False-Positive Control, Human-In-The-Loop, Latency, gent Tracing & Observability (LangSmith), Token & Cost Tracking, Latency Benchmarking.
- **Data Engineering & Cloud**: Python, PySpark, SQL, Java, FastAPI, Databricks (Auto Loader, DLT, Lakeflow, Medallion Architecture), Snowflake (Snowpipe, Streams & Tasks, Snowsight), MySQL, AWS S3/EC2, MLflow, AutoML, Git/Bitbucket.
- **Solution Engineering**: Technical Feasibility Analysis, Product Gap Analysis, Enterprise Application Integration, Proofs Of Concept (POCs), Solution Delivery, Product Management, Stakeholder Management, Technical Documentation.

ACCOMPLISHMENT

- **Certifications**: Model Context Protocol ([Anthropic](#)), Claude with Amazon Bedrock ([Anthropic](#)), Claude with Google Cloud's Vertex AI ([Anthropic](#)), Building with Claude API ([Anthropic](#)), Introduction to agent skills (Anthropic), Introduction to subagents (Anthropic), Snowflake Data Engineering ([Snowflake](#)), McKinsey Forward Learning ([Mckinsey.org](#)).
- **Recognition**: Exceptional Performer Award (UKG -2022), Technical Excellence Awards (UKG -2021).
- **Leadership**: Graduate Assistant (USF-2024), Data Science Club (USF-2023), PMI Volunteer (Tampa-2022).